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Introduction: 44

Machine Learning (ML), Artificial Intelligence (AI), Neural Networks (NN), Augmented Reality (AR), Virtual Reality (VR), and Mixed Reality (MR) are key technologies enhancing data visualization in healthcare. This evaluation examines recent advancements and their impact on visualizing healthcare data. 44

1. Machine Learning (ML): 44

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Task 1

- **Introduction**

Data Visualization uses charts, graphs, and maps to make complex data more accessible and actionable. It highlights patterns, trends, and insights, aiding quick, informed business decisions.

Grammar and Idioms in Data Visualization guide how data is visually represented. "Grammar" involves selecting the right visual elements, while "Idioms" are standard patterns, like line graphs for trends or bar charts for comparisons. Together, they ensure data tells a clear story, making it easy to understand without extra explanations.

Grammar in Data Visualization

- **Definition and Importance:**

In data visualization, 'grammar' refers to the rules and conventions that govern the construction of visual representations of data. This includes the use of elements such as axes, scales, labels, colors, and shapes to convey information effectively and accurately. Just as grammatical rules in language ensure clarity and comprehension, the grammar of data visualization ensures that visualizations are interpretable, consistent, and communicative.

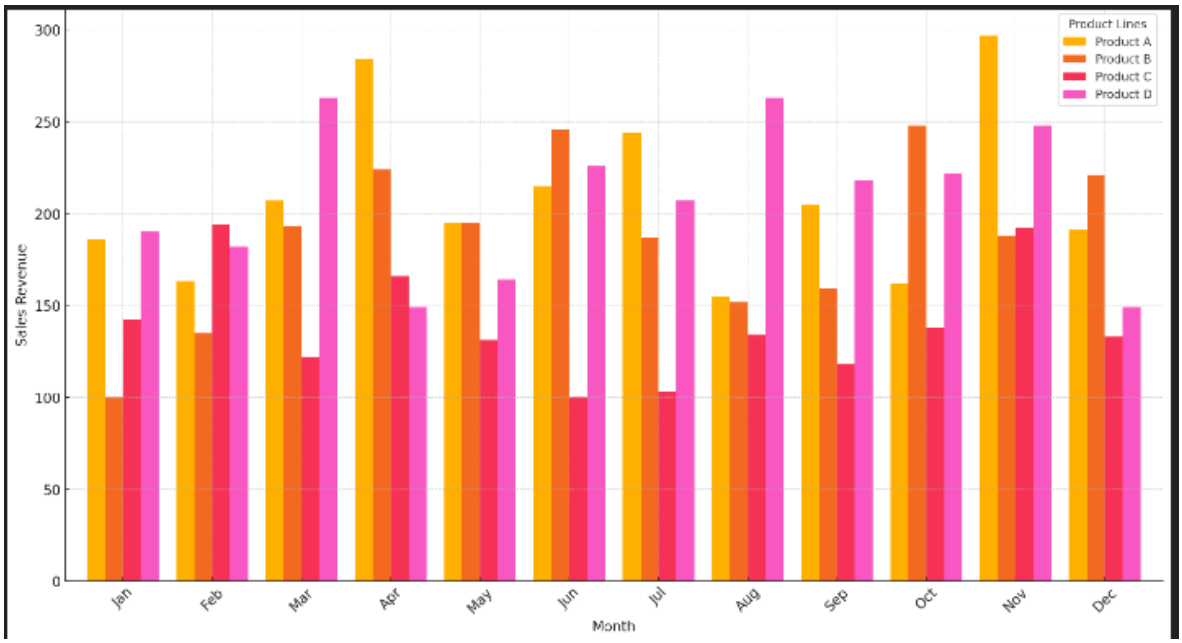
- **Examples:**

➢ **Bar Charts and Histograms:**

- **Bar Charts:**

Bar charts compare quantities across categories, with each bar's height representing the value. The x-axis lists categories, and the y-axis shows their values, enabling quick comparisons.

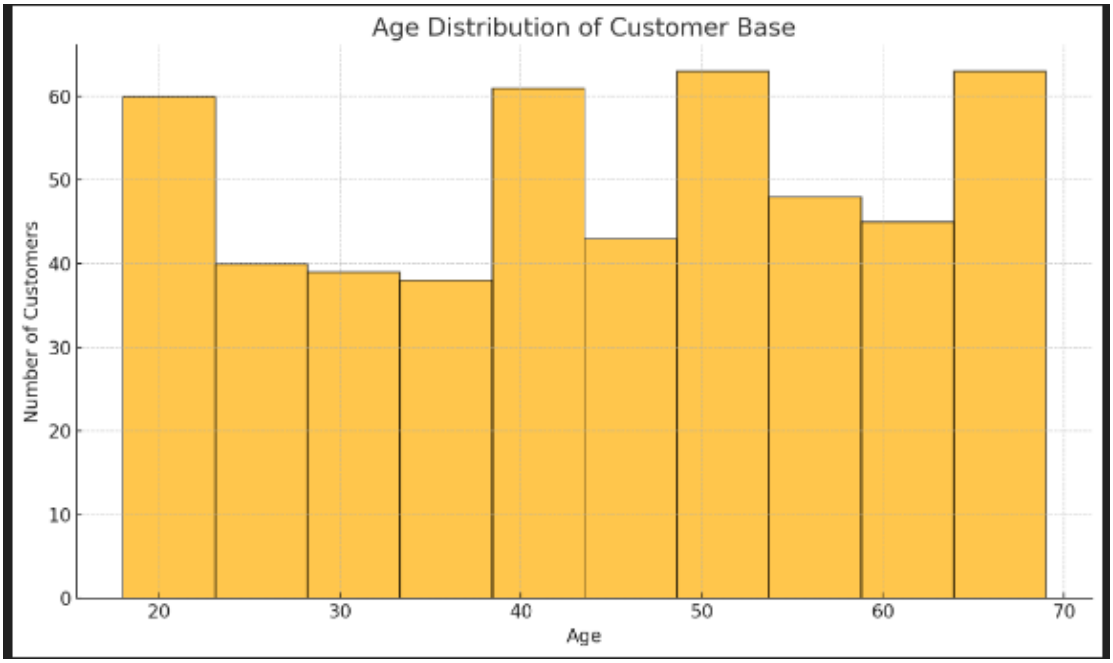
Example: A bar chart comparing monthly sales across different product lines can show which product lines are performing better or worse relative to each other.



- **Histograms:**

Histograms use grammar to show frequency distributions. The x-axis represents continuous intervals of data, and the y-axis shows the frequency of data points within each interval. This helps in understanding the distribution and spread of data.

Example: A histogram of customer ages can reveal the age distribution of a company's customer base, highlighting which age groups are most common.



- **Line Graphs:**

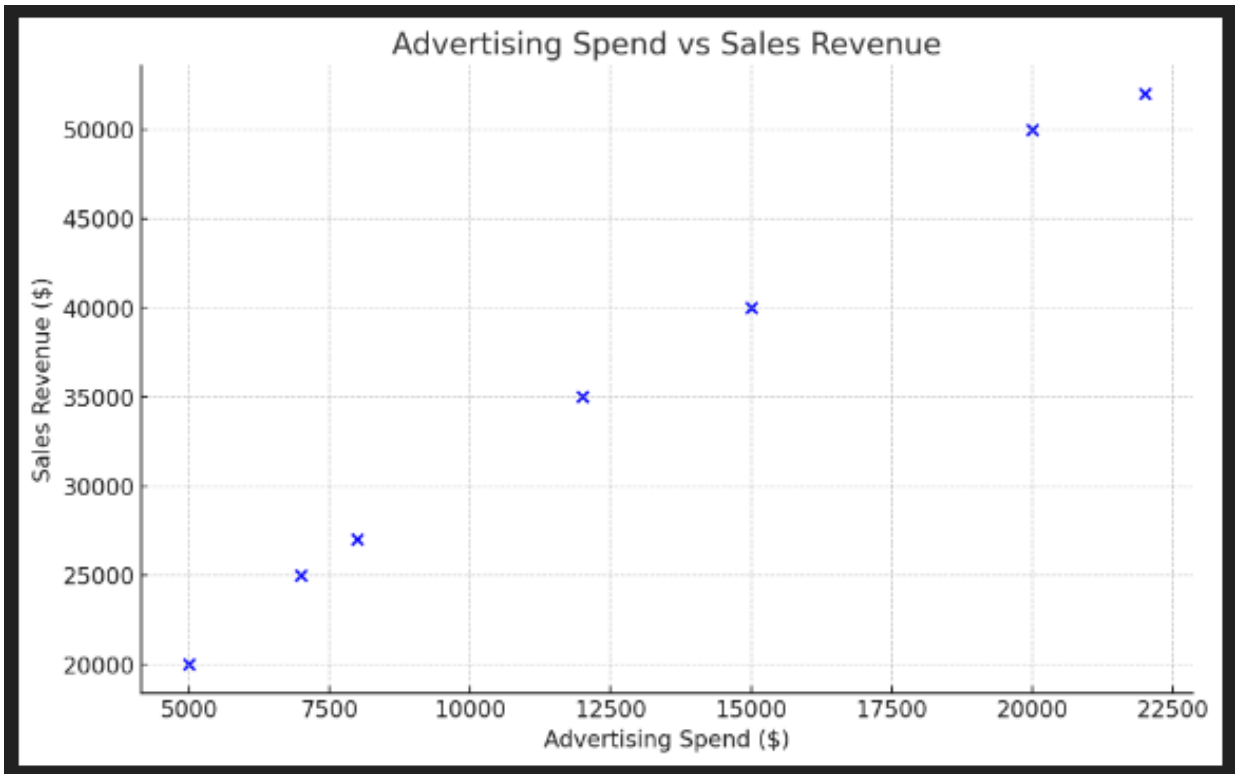
Line graphs display trends over time by connecting data points on a two-dimensional plane, with time on the x-axis and the measured variable on the y-axis. This format highlights changes and trends over time.

Example: A line graph showing monthly revenue over the past year can highlight periods of growth or decline and help identify seasonal patterns.

- **Scatter Plots:**

Scatter plots show relationships between variables, with each point representing an observation. The x and y axes represent different variables, helping identify correlations, clusters, and outliers.

Example: A scatter plot showing advertising spend versus sales revenue can reveal whether there is a correlation between the two variables, indicating how advertising impacts sales.



- **Benefits:**

Proper syntax in data visualization clarifies trends and comparisons, reducing misunderstandings and enhancing decision-making. Consistent visuals build trust and ensure effective storytelling.

❖ Idioms in Data Visualization

- **Definition and Importance:**

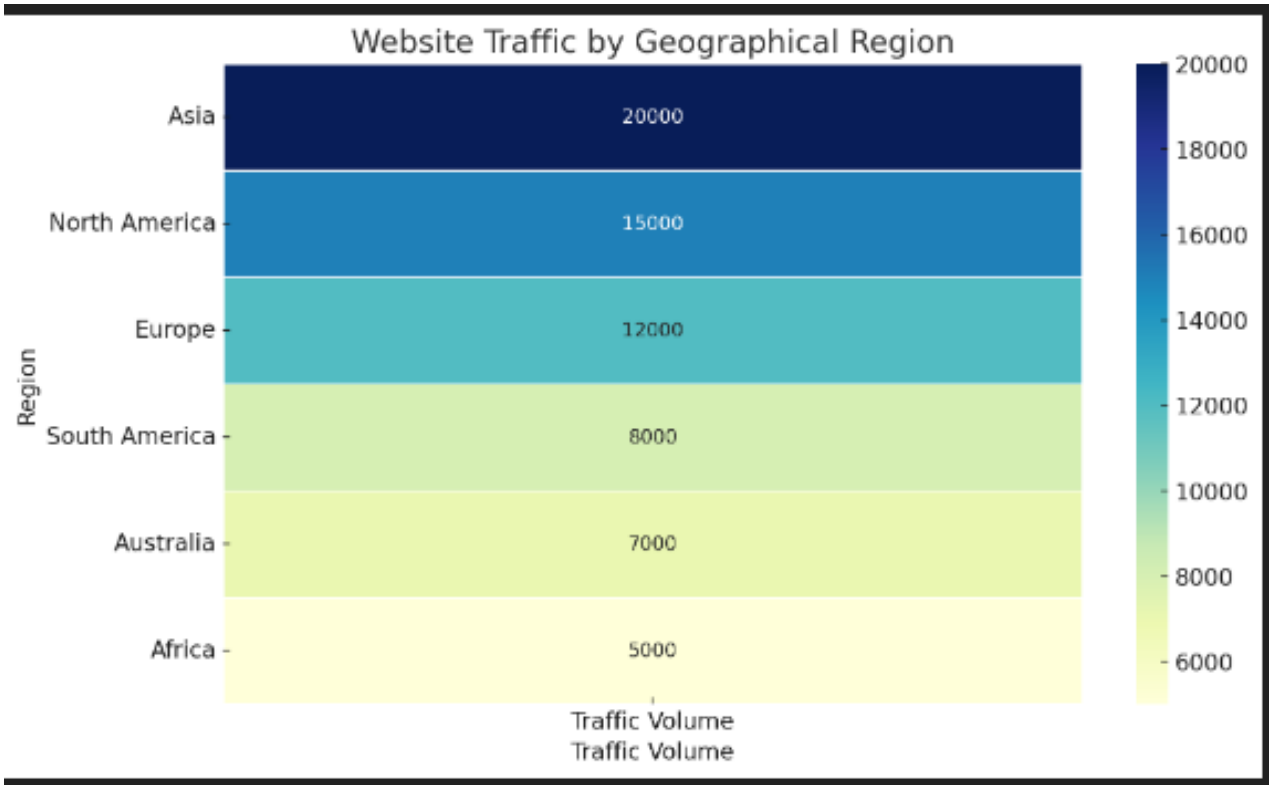
'Idioms' in data visualization are standard graphical representations that use familiar visual metaphors for easy recognition and understanding, enhancing quick comprehension and effective communication of complex data.

- **Examples:**

➢ **Heat Maps:**

✦ **Usage:** Heat maps use color intensity to represent data, allowing quick identification of patterns and outliers.

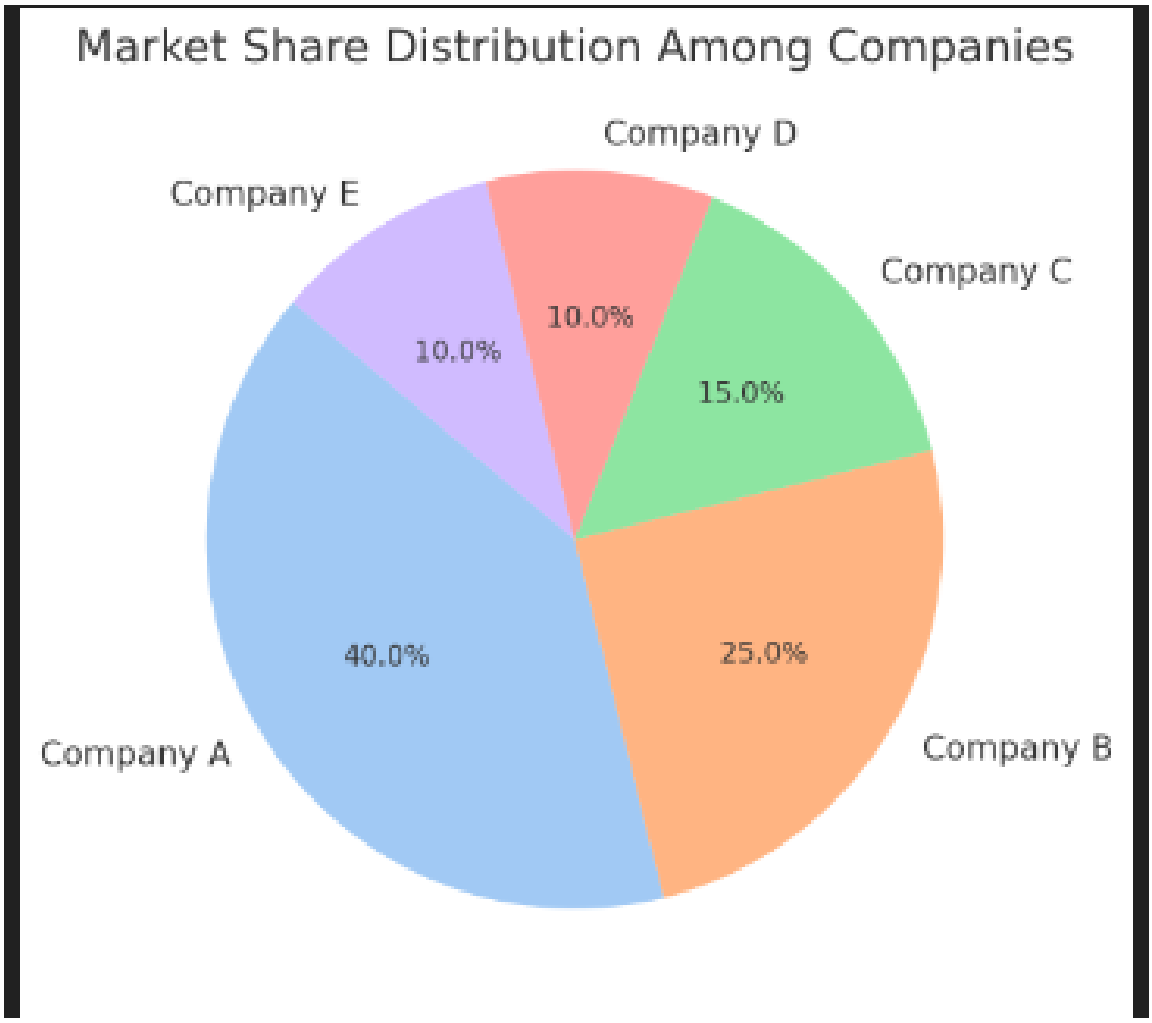
✦ **Example:** In a heat map of website traffic by region, darker shades indicate higher traffic, helping businesses quickly spot high-traffic areas and potential markets.



- **Pie Charts:**

- **Usage:** Pie charts use segments to represent parts of a whole, with each segment's size showing its proportion of the total, making it easy to see relative sizes.

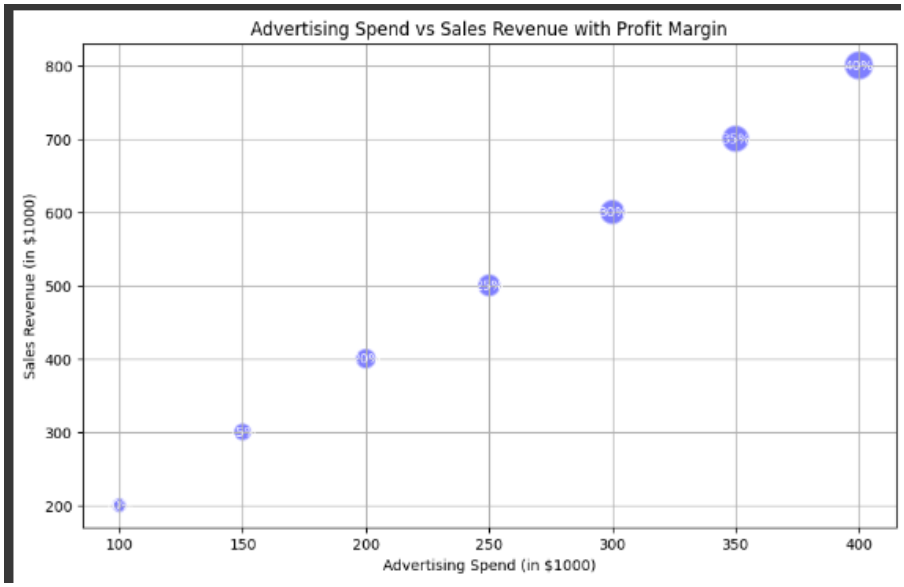
- **Example:** A pie chart shows how companies divide market share, making it easy to see each company's position. For example, if Company A holds 40% of the market, its segment will cover 40% of the chart, illustrating its proportion.



- **Bubble Charts:**

- **Usage:** Bubble charts use the size of bubbles to represent a third variable, adding another dimension to data understanding. The position of the bubble on the x and y axes represents two variables, while the bubble size indicates a third variable.

- **Example:** A bubble chart shows the relationship between advertising spend (x-axis) and sales revenue (y-axis), with bubble size representing profit margin. Larger bubbles indicate higher profit margins, offering a multi-dimensional view of how advertising affects revenue and profit.



- **Benefits:**

Data visualization idioms reduce cognitive load, enabling quick insights and faster decision-making. Familiar idioms help everyone easily interpret data and make informed strategic choices.

Application in Business Context

- **Application to SWOT Analysis Using Data Visualization**

In the context of SWOT analysis, data visualizations can provide clear and immediate insights into a company's strengths, weaknesses, opportunities, and threats. Here's how different types of visualizations can be applied:

1. **Strengths: Bar Graph Highlighting Strong Sales Performance**

Scenario:

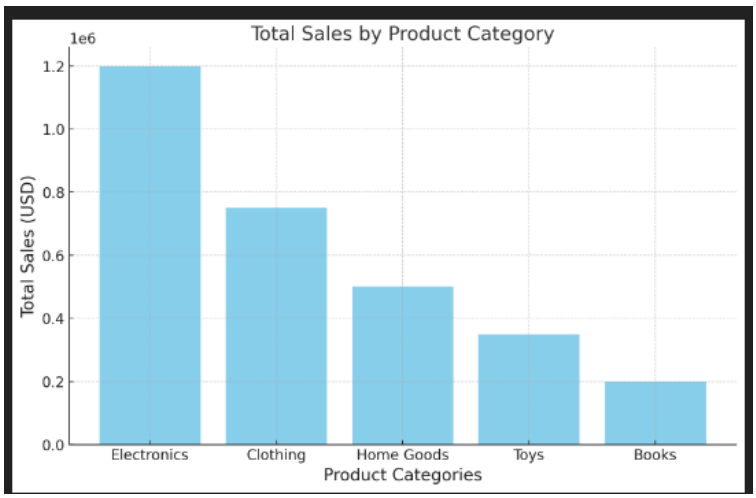
- A retail company wants to understand which of its product categories are performing the best in terms of sales.

Visualization:

- Bar Graph Setup: The x-axis represents different product categories (Electronics, Clothing, Home Goods), and the y-axis represents the total sales in USD for each category.
- Data Representation: Each bar corresponds to a product category, with the height of the bar indicating the total sales for that category.

Insights:

- The bar graph highlights strong sales in Electronics, showing a strength in that segment and making performance comparisons across categories clear.



2. Weaknesses: Line Chart Showing a Declining Trend in Customer Retention

Scenario:

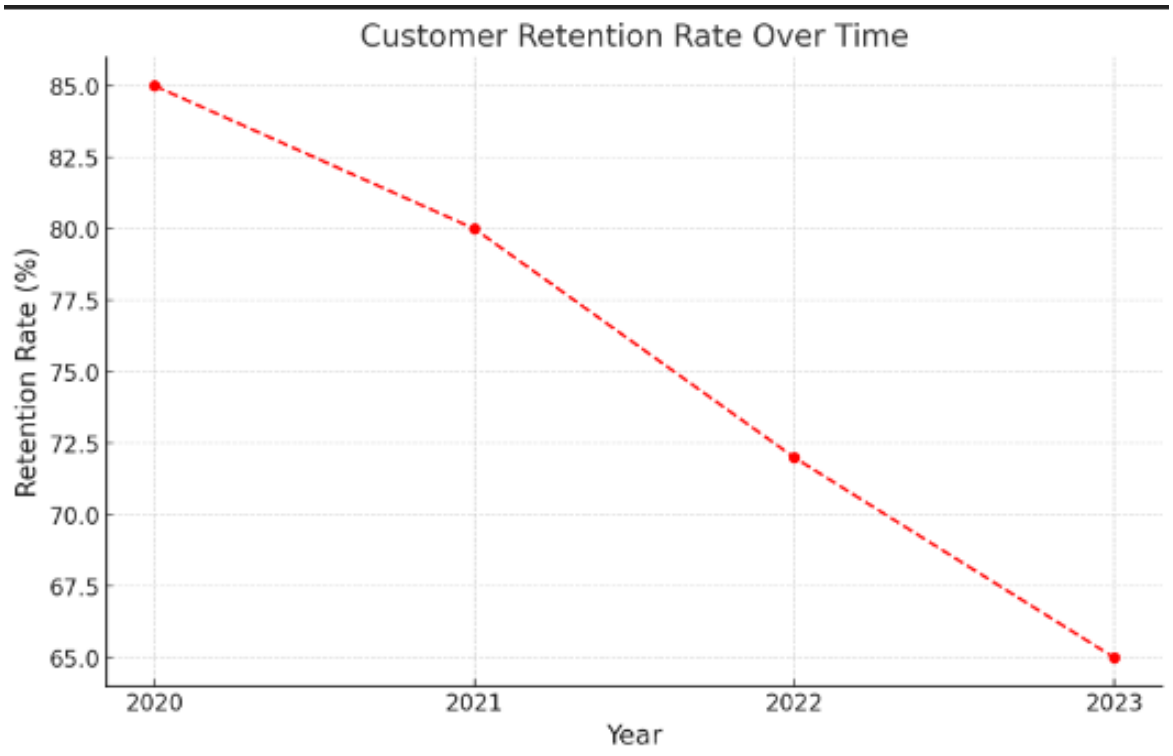
- A company is concerned about a potential decline in customer loyalty and wants to analyze customer retention rates over the past few years.

Visualization:

- Line Chart Setup: The x-axis represents the timeline (quarters or years), and the y-axis represents the customer retention rate as a percentage.
- Data Representation: A line is plotted to show the retention rate over time.

Insights:

- The line chart reveals a steady decline in retention rate over two years, indicating a weakness in customer engagement. The trend is clear, urging further investigation and corrective action.



3. Opportunities: Scatter Plot Identifying Market Gaps

Scenario:

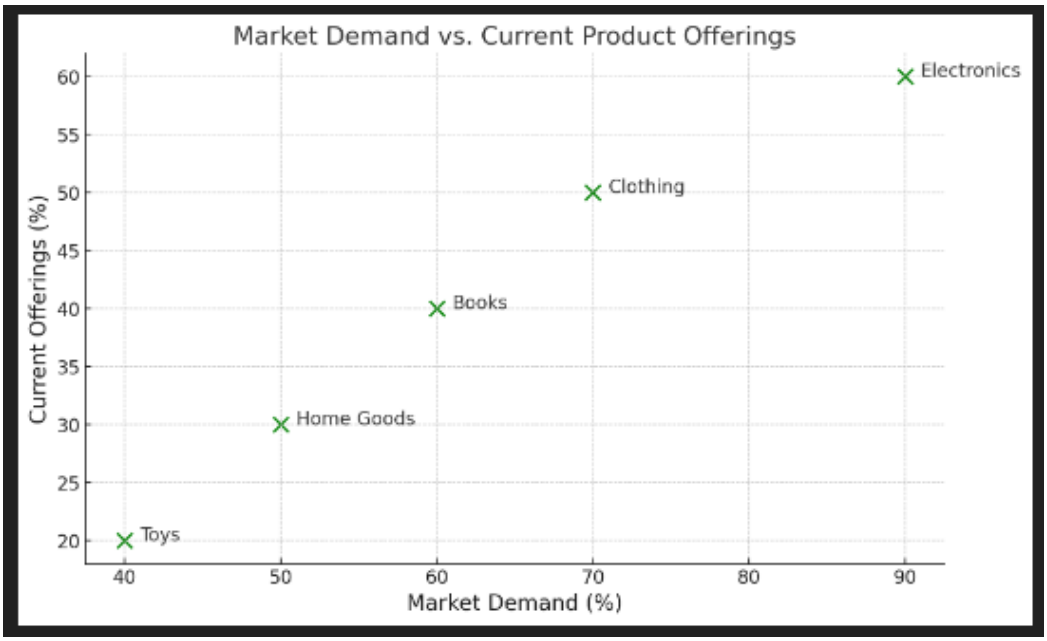
- A company wants to explore new market opportunities by analyzing the relationship between market demand and current product offerings.

Visualization:

- Scatter Plot Setup: The x-axis represents market demand for different product categories, and the y-axis represents the company's current product offerings.
- Data Representation: Each point on the scatter plot represents a product category, with its
- position indicating the level of demand and current market saturation.

Insights:

- The scatter plot shows market gaps in high-demand product categories, indicating opportunities for growth. Expanding offerings in these areas can help capture new opportunities.



4. Threats: Radar Chart Showing Competitor Performance

Scenario:

- A company wants to compare its performance against key competitors across multiple critical business metrics, market share, product quality, customer satisfaction).

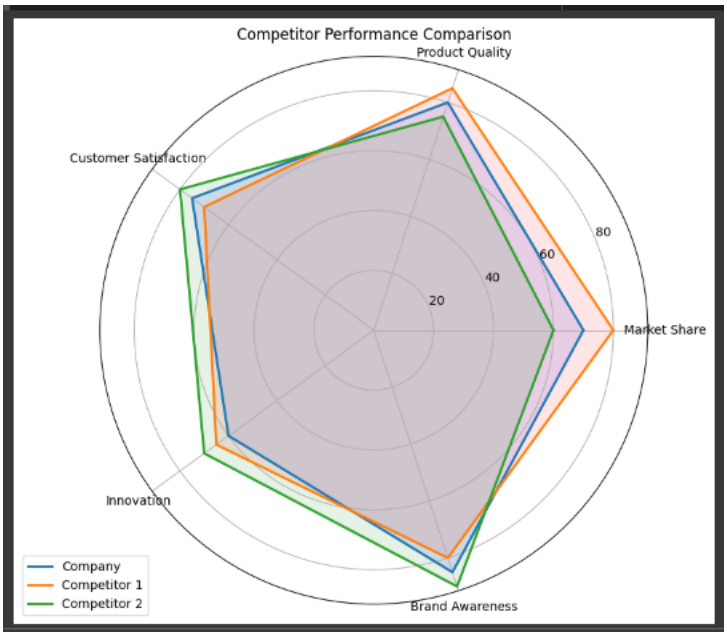
Visualization:

- Radar Chart Setup: Each axis of the radar chart represents a different metric, with the performance of the company and its competitors plotted on the same chart.

- Data Representation: The company's performance and that of its competitors are displayed as different shapes on the radar chart, allowing for direct comparison across all metrics.

Insights:

- The radar chart shows competitors outperforming in areas like customer satisfaction and market share, highlighting threats. It offers a clear comparison, guiding strategic improvements to mitigate these risks.



❖ Conclusion

In summary, grammar and idioms are essential for clear and effective data visualizations. Proper grammar ensures clarity, while familiar idioms help viewers quickly understand the information. Effective use of these elements aids businesses in identifying strengths, weaknesses, opportunities, and threats, leading to better decision-making.

Task 2

- **Selected Domain of Interest: Healthcare**

Medical technology generates vast, diverse data daily, crucial for strategic planning, operations, and patient care. Analyzing data at all levels reveals key business insights and drives decision-making.

- **Types of Data Generated in Healthcare**

1. Structured Data:
Structured data in healthcare is highly organized and easily searchable, typically found in databases. Examples include:

- **Electronic Health Records:** Patient information such as demographics, medical history, diagnoses, medications, immunization dates, allergies, radiology images, and laboratory test results.

- **Billing and Claims Data:** Information related to patient billing, insurance claims, payments, and coding for medical procedures and diagnoses.

2. Semi-Structured Data:
Semi-structured data in healthcare has some organizational properties but is not fully structured. Examples include:

- **Emails and Communication Logs:** Correspondence between healthcare providers, staff, and patients, often containing a mix of structured elements and unstructured text.

- **Clinical Notes:** Written observations by healthcare providers that are partly structured but also contain unstructured narrative text.

2. **Unstructured Data:**

Unstructured data in healthcare lacks a predefined structure, making it harder to analyze. Examples include:

- **Medical Imaging:** Radiology images, pathology slides, and other visual data used for diagnosis and treatment planning.
- **Patient Surveys and Feedback:** Text responses from patient satisfaction surveys, online reviews, and feedback forms.

Role of Data in Decision-Making in Healthcare

1. Operational Decisions:

- Operational decisions in healthcare rely on real-time data for efficient patient care and hospital management. Examples include:
- **Patient Care Management:** Real-time monitoring of patient vitals and clinical status using EHR data, allowing healthcare providers to make immediate adjustments to treatment plans.
- **Resource Allocation:** Using real-time data to manage the availability of medical staff, hospital beds, and medical equipment to meet patient needs and optimize workflow.

2. Strategic Decisions:

Strategic decisions are long-term, guided by data trends, shaping healthcare direction, policies, and investments. Examples include:

- **Population Health Management:** Analyzing patient data to identify trends in health outcomes, disease prevalence, and risk factors. This information helps in developing public health initiatives and preventive care programs.
- **Healthcare Policy and Planning:** Using data on healthcare utilization, costs, and outcomes to inform policy decisions, allocate funding, and plan for future healthcare needs.

3. Tactical Decisions:
Tactical decisions are short-term actions that support strategic goals, aiming to achieve objectives quickly. Examples include:

- **Targeted Health Campaigns:** Using patient data to identify high-risk populations and design targeted health campaigns, such as vaccination drives or chronic disease management programs.

- **Financial Management:** Utilizing billing and claims data to optimize revenue cycle management, identify cost-saving opportunities, and ensure compliance with regulatory requirements.

Critically Review the Lowest Grain of Data Collected in Healthcare

The Smallest Unit of Data: Individual Patient Encounter Record

A medical record is often the smallest data unit in healthcare, detailing a single patient-provider encounter. Key elements may include:

- Patient Identification: Unique patient ID, name, date of birth, gender.
- Encounter Details: Date and time of visit, type of visit.
- Vital Signs: Blood pressure, heart rate, temperature, respiratory rate.
- Clinical Observations: Symptoms reported, physical examination findings.
- Diagnosis: Medical conditions diagnosed during the visit.
- Treatment and Procedures: Medications prescribed, treatments administered, surgical procedures performed.
- Provider Information: Identifying information about the healthcare provider involved.

Granularity and Aggregation

- **Granularity:**

Granularity refers to the level of detail in the data. The individual patient encounter record is highly granular, capturing specific details about each patient's visit. This granularity is crucial for providing personalized care and accurate medical records.

- **Aggregation:**

Individual patient encounter records can be aggregated in various ways to generate meaningful insights and support decision-making:

1.Patient-Level Aggregation:

- **Patient History:** Aggregating all encounter records for a single patient to provide a comprehensive medical history. This helps in continuity of care and informed treatment decisions.

2. Department-Level Aggregation:

- **Department Performance:** Summarizing data by department to evaluate performance metrics such as patient outcomes, average length of stay, and resource utilization.

3. Facility-Level Aggregation:

- **Operational Efficiency:** Aggregating data across the entire healthcare facility to monitor key performance indicators such as bed occupancy rates, average wait times, and overall patient satisfaction.

4. Population-Level Aggregation:

- **Public Health Insights:** Combining data from multiple facilities to analyze population health trends, disease prevalence, and risk factors. This supports public health initiatives and policymaking.

✦ **Example of Aggregation and Insights**

➢ **Example: Hospital Readmissions**

- Granular Data: Individual patient encounter records including discharge information and subsequent readmissions.

- Aggregation: Analyzing readmission rates by patient demographics, diagnosis, and treatment protocols.

- Insight: Identifying patterns and risk factors associated with high readmission rates. This insight can lead to targeted interventions to reduce readmissions, such as enhanced discharge planning and follow-up care.

➢ **Example: Medication Usage**

- **Granular Data:** Detailed records of medications prescribed during each patient encounter.

- **Aggregation:** Summarizing medication usage across different departments and patient populations.

- **Insight:** Identifying trends in medication prescriptions, potential overuse of certain drugs, and opportunities for cost savings through formulary management.

❖ Rolling Up Granular Data to Summarized Categories in Healthcare

In healthcare, various aggregation and categorization methods can group patient data into summary categories, aiding strategic planning and decision-making by providing clear, consolidated insights.

1. Aggregation

- Aggregation involves summarizing data points to provide a higher-level view. Examples in healthcare include:

❖ Patient Volume:

- **Daily/Monthly Patient Visits:** Summarizing the total number of patient visits per day, week, or month.
- **Insight:** Helps in understanding patient flow, peak times, and resource allocation needs.

Clinical Outcomes:

- **Treatment Success Rates:** Aggregating data on treatment outcomes across different conditions and interventions.
- **Insight:** Identifies effective treatments and areas needing improvement.

Financial Performance:

- **Revenue and Costs:** Summarizing billing and claims data to calculate total revenue and operational costs.
- **Insight:** Assists in financial planning and identifying cost-saving opportunities.

2. Categorization

- Categorization involves grouping data into categories for more detailed analysis. Examples include:

Patient Demographics:

- **Age, Gender, Ethnicity:** Grouping patients by demographic characteristics.
- **Insight:** Reveals demographic-specific health trends and needs, guiding targeted healthcare programs.

Disease Categories:

- **Chronic vs. Acute Conditions:** Categorizing patient encounters based on disease types.
- **Insight:** Helps in resource planning and developing specialized care protocols.

Service Utilization:

- **Inpatient vs. Outpatient Services:** Grouping encounters by type of service provided.
- **Insight:** Assists in understanding service demand and optimizing facility usage.

3. Trends and Patterns:

- Trends and patterns analysis involves identifying changes and recurring themes over time. Examples include:

❖ **Disease Incidence Trends:**

- **Seasonal Variations:** Tracking the incidence of diseases over different seasons.
- **Insight:** Helps in preparing for seasonal outbreaks and allocating resources accordingly.

❖ **Patient Behavior Patterns:**

- **Healthcare Utilization:** Analyzing how often patients seek medical care and for what reasons.
- **Insight:** Identifies trends in patient behavior, informing patient engagement and preventive care strategies.

❖ **Financial Trends:**

- **Revenue Cycles:** Analyzing revenue patterns over time.
- **Insight:** Helps in identifying financial peaks and troughs, enabling better budgeting and financial management.

❖ **Examples of Rolling Up Data to Generate Business Insights**

✦ **Example 1: Hospital Readmission Rates**

- **Granular Data:** Individual readmission records, including patient demographics, initial diagnoses, and treatment details.
- **Aggregation:** Calculate the total number of readmissions per month.
- **Categorization:** Group readmissions by diagnosis.
- **Trends and Patterns:** Analyze readmission rates over time to identify spikes.

o **Insight:** Understanding the patterns and causes of readmissions can lead to targeted interventions to reduce readmission rates, improving patient outcomes and reducing costs.

❖ **Generating Business Insights from Summarized Data in Healthcare**

✦ Summarized data categories in healthcare can be analyzed to generate actionable business insights through descriptive, predictive, and prescriptive analytics. Here's how each type of analytics can provide value:

1. Descriptive Analytics: Understanding What Happened

- Descriptive analytics focuses on analyzing historical data to understand past events.

✦ **Example: Monthly Patient Visit Reports**

- **Data Summarized:** Total number of patients visits each month.
- **Insight:** Identify which months had higher or lower patient volumes.
- **Action:** Adjust staffing levels and resources based on peak and off-peak periods to improve efficiency and patient care.

2. Predictive Analytics: Forecasting Future Trends

- Predictive analytics uses historical data to predict future events.

✦ **Example: Predicting Disease Outbreaks**

- **Data Summarized:** Trends in disease incidence over the past years.
- **Insight:** Forecast potential outbreaks of diseases like the flu based on historical patterns.
- **Action:** Prepare for predicted outbreaks by stocking up on vaccines, increasing public health campaigns, and allocating necessary medical staff and resources.

3. Prescriptive Analytics: Recommending Actions

- Prescriptive analytics suggests actions to take based on data insights.

✦ Example: Optimizing Medication Inventory

- **Data Summarized:** Medication usage rates and inventory levels.
- **Insight:** Identify which medications are used most frequently and predict future demand.
- **Action:** Recommend ordering schedules and inventory levels to ensure critical medications are always in stock without overstocking, reducing costs, and minimizing waste.

Task 3

Introduction

Healthcare generates vast data daily, impacting lives and well-being. Data visualization simplifies complex information into actionable insights, aiding in trend recognition, decision-making, and improving patient outcomes. Effective visualizations enhance resource use, optimize processes, and improve clinical results.

- **Data preparation**

The dataset I used for this project is a comprehensive healthcare dataset that includes various key fields essential for analyzing patient care, hospital operations, and financial aspects. The key fields in this dataset are:

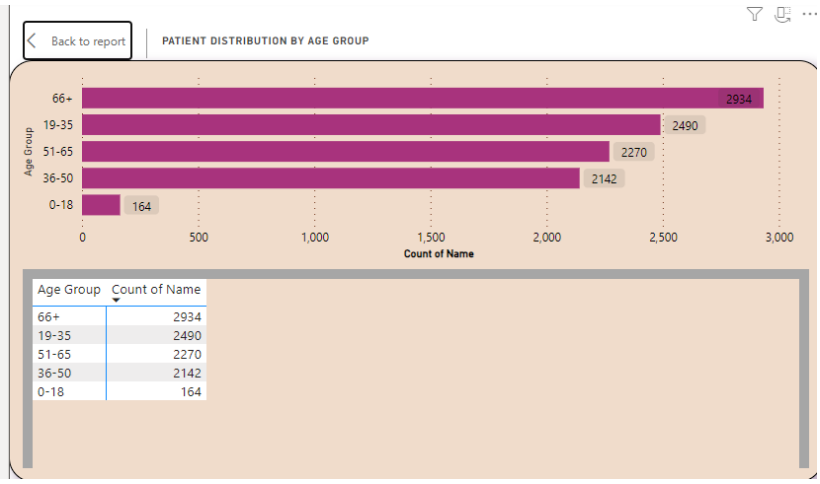
- Name: The full name of the patient.
- Age: The age of the patient at the time of admission.
- Gender: The gender of the patient (Male, Female).
- Blood Type: The blood type of the patient
- Medical Condition: The primary medical condition or diagnosis of the patient.
- Date of Admission: The date on which the patient was admitted to the hospital.
- Doctor: The name of the doctor responsible for the patient's care.
- Hospital: The hospital where the patient was admitted.
- Insurance Provider: The name of the insurance company covering the patient's medical expenses.
- Billing Amount: The total amount billed for the patient's treatment.
- Room Number: The hospital room number where the patient was accommodated.
- Admission Type: The type of admission (Emergency, Elective).
- Discharge Date: The date on which the patient was discharged from the hospital.
- Medication: The list of medications prescribed to the patient during their stay.
- Test Results: The results of any medical tests conducted during the patient's stay.

➢ **Dashboard Creation**

❖ **Patient Demographics Overview**

1. Patient Distribution by Age Group (Bar Chart):

- This bar chart visually represents the distribution of patients across various age groups, helping to identify the age demographics most frequently admitted.



× **Explanation:**

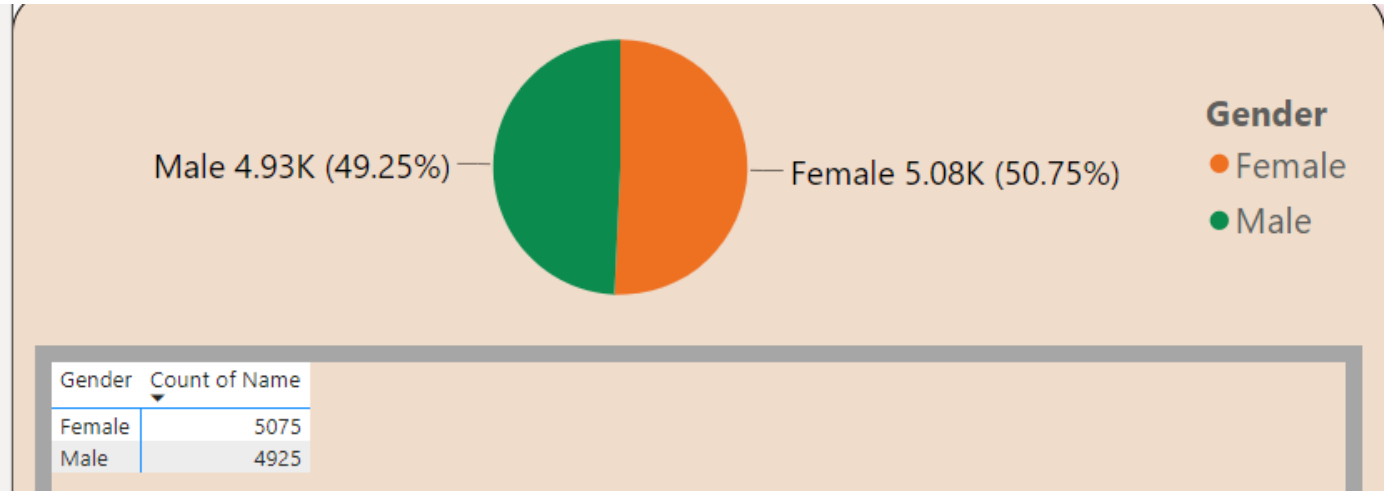
- 66+ years: The largest group, indicating a significant elderly population needing medical attention.
- 19-35 years: The second-largest group, reflecting common healthcare needs among young adults.
- 51-65 years and 36-50 years: Middle-aged groups with substantial numbers, likely requiring chronic care or preventive services.
- 0-18 years: The smallest group, indicating fewer hospital visits or admissions for children.

× **Decision:**

- Resource Allocation: Prioritize resources, staffing, and specialized care for the 66+ age group due to their higher healthcare demands.
- Targeted Programs: Develop specific healthcare programs for young adults and middle-aged groups to address common health issues.
- Pediatric Care: Though smaller, ensure pediatric care remains available and focused on preventive care and early intervention for the 0-18 age group.

2.Gender Distribution (Pie Chart):

- The Gender Distribution Pie Chart shows that out of a total of 10,000 patients, 5,075 are female and 4,925 are male. This results in a near-equal distribution, with females making up 50.75% of the population and males accounting for 49.25%.



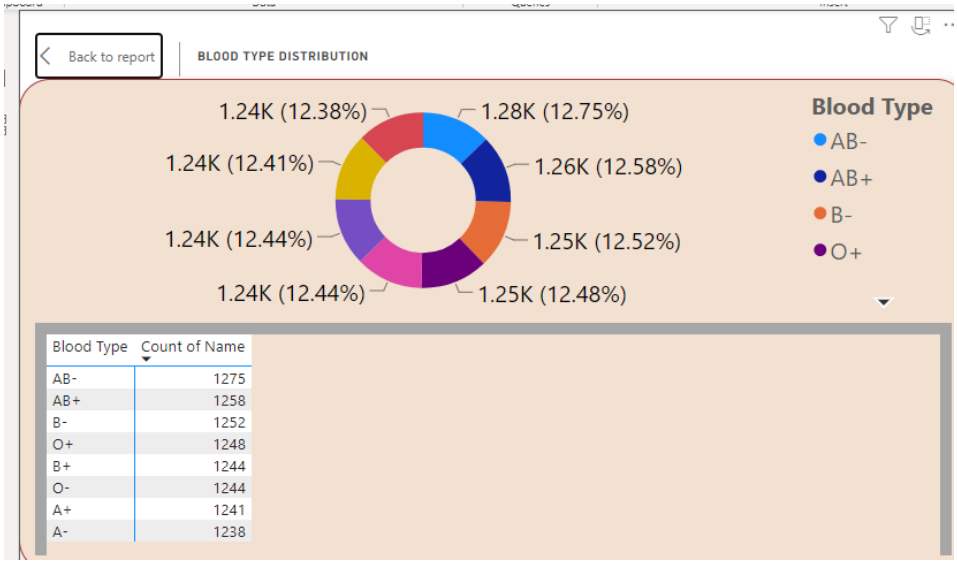
Gender	Count of Name
Female	5075
Male	4925

- Female Patients: 5,075 (50.75%)
- Male Patients: 4,925 (49.25%)

3.Blood Type Distribution (Donut Chart):

The Blood Type Distribution Donut Chart shows the distribution of different blood types within patient population. The values are very close, indicating that each blood type is represented in similar proportions.

- **AB-**: 1,275 (12.75%)
- **AB+**: 1,258 (12.58%)
- **B-**: 1,252 (12.52%)
- **O+**: 1,248 (12.48%)
- **B+**: 1,244 (12.44%)
- **O-**: 1,244 (12.44%)
- **A+**: 1,241 (12.41%)
- **A-**: 1,238 (12.38%)



Explanation:

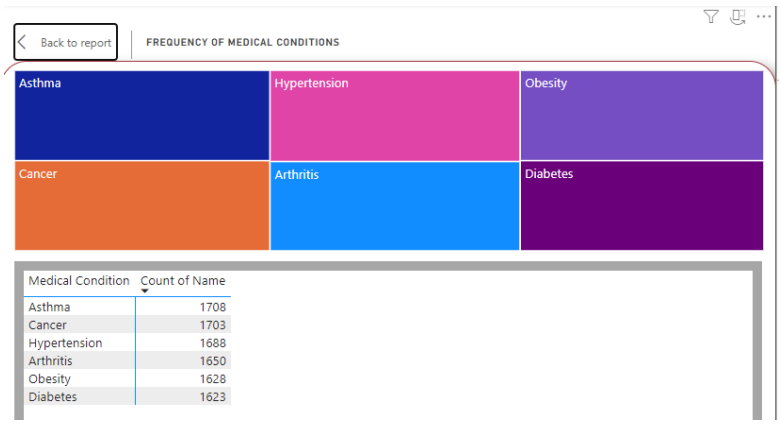
- The distribution of blood types is uniform, with each type representing around 12-13% of the total population.
- The smallest difference between blood types is only 37 patients, indicating no significant concentration of any blood type.
- This prevents shortages and ensures that all patients, regardless of blood type, have access to necessary transfusions.

❖ **Medical Condition and Treatments Section:**

1.Frequency of Medical Conditions (Tree Map):

The Frequency of Medical Conditions Tree Map shows the distribution of various medical conditions within the patient population.

- **Asthma:** 1,708 (17.08%)
- **Cancer:** 1,703 (17.03%)
- **Hypertension:** 1,688 (16.88%)
- **Arthritis:** 1,650 (16.50%)
- **Obesity:** 1,628 (16.28%)
- **Diabetes:** 1,623 (16.23%)



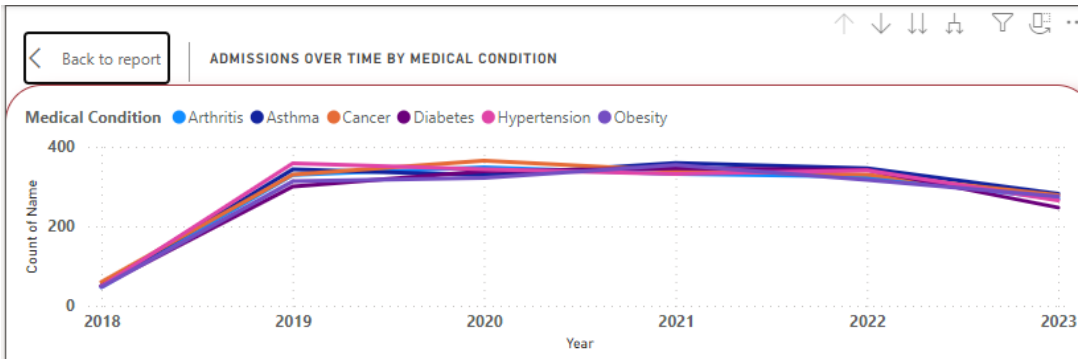
Explanation:

- The distribution of medical conditions is balanced, with each condition representing a similar percentage of the population.
- The most common condition is Asthma (17.08%), closely followed by Cancer (17.03%) and Hypertension (16.88%).
- Diabetes (16.23%) is the least common among these conditions, but the difference is minimal, indicating that all these conditions are prevalent at similar rates.

2.Admissions Over Time by Medical Condition (Line Chart):

Explanation:

The line chart titled "Admissions Over Time by Medical Condition" shows the trend of patient admissions for various medical conditions from 2018 to 2023. The conditions tracked are Arthritis, Asthma, Cancer, Diabetes, Hypertension, and Obesity. Here's what the chart indicates:



Year	Arthritis	Asthma	Cancer	Diabetes	Hypertension	Obesity
2018	47	50	60	50	49	47
2019	329	343	330	300	358	313
2020	348	328	365	338	343	322
2021	332	359	340	346	332	354
2022	325	346	330	342	341	317
2023	269	282	278	247	265	275

Trends Over Time:

- 2018-2020: There is a noticeable increase in admissions across all medical conditions, peaking around 2020.
- 2021-2023: After 2020, admissions for all conditions slightly decline but remain relatively stable with a gentle downward trend.

Conditions Comparison:

- All the conditions follow a similar pattern in terms of admissions, with no single condition showing a dramatic divergence from the others.
- Asthma and Hypertension seem to have a higher count of admissions compared to other conditions during the peak years (2019-2020).

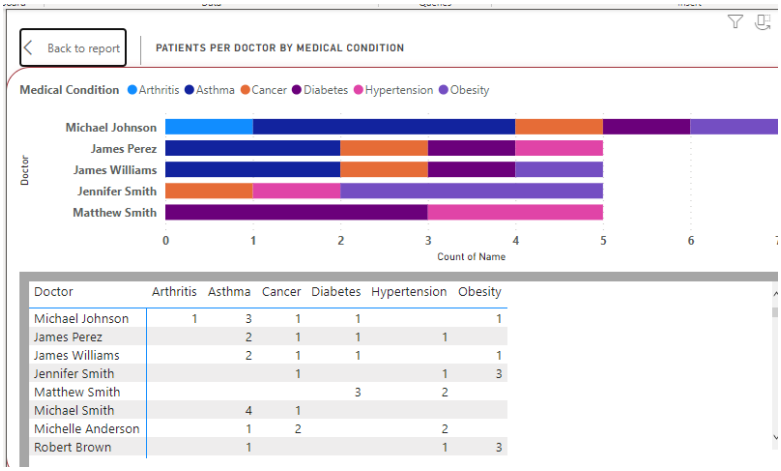
Latest Year (2023):

- By 2023, admissions have decreased from their peak in 2020 but remain significant, with all conditions still having over 200 admissions.

3.Patients Per Doctor by Medical Condition (Stacked Bar Chart):

Explanation:

The "Patients Per Doctor by Medical Condition" stacked bar chart shows the number of patients each doctor treats, categorized by conditions like Arthritis, Asthma, Cancer, Diabetes, Hypertension, and Obesity.



Doctor	Arthritis	Asthma	Cancer	Diabetes	Hypertension	Obesity
Michael Johnson	1	3	1	1		1
James Perez		2	1	1	1	
James Williams		2	1	1		1
Jennifer Smith			1		1	3
Matthew Smith				3	2	
Michael Smith		4	1			
Michelle Anderson		1	2		2	
Robert Brown		1			1	3

Doctor Workload:

- Michael Johnson: Focuses on Asthma and Cancer.
- James Perez: Handles all conditions, with emphasis on Obesity and Asthma.
- James Williams: Primarily treats Cancer and Diabetes.
- Jennifer Smith: Manages more patients with Hypertension and Obesity.
- Matthew Smith: Focuses on Obesity, Hypertension, and Cancer.

Condition Distribution:

- Obesity and Hypertension are most common, especially for Jennifer and Matthew Smith.
- Cancer and Asthma are significant, mainly for Michael Johnson and James Williams.

Patient Distribution:

- The patient load varies slightly by doctor, with some handling more patients based on their specialization.

Decision: Balance Workload

- Redistribute Patients: If doctors like Jennifer Smith and Matthew Smith have too many patients with Obesity and Hypertension, reassign some patients to other doctors to balance the workload.

❖ **Financial Data Section:**

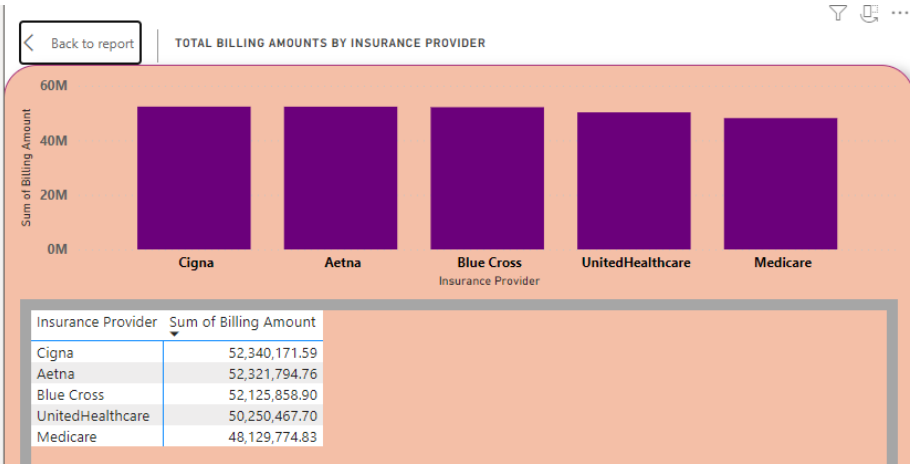
1.Total Billing Amounts by Insurance Provider (Bar Chart):

The bar chart at the top represents the sum of billing amounts for five different insurance providers: Cigna, Aetna, Blue Cross, UnitedHealthcare, and Medicare.

Data Table:

- Below the chart, there's a table listing the insurance providers along with their corresponding sum of billing amounts.

- The values are:
  - **Cigna**: $52,340,171.59
  - **Aetna**: $52,321,794.76
  - **Blue Cross**: $52,125,858.90
  - **UnitedHealthcare**: $50,250,467.70
  - **Medicare**: $48,129,774.83



Insurance Provider	Sum of Billing Amount
Cigna	52,340,171.59
Aetna	52,321,794.76
Blue Cross	52,125,858.90
UnitedHealthcare	50,250,467.70
Medicare	48,129,774.83

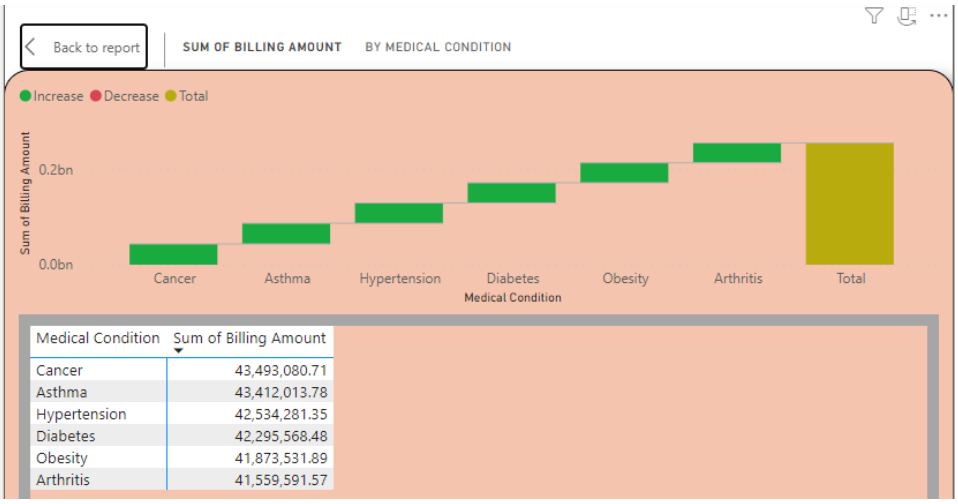
Explanation:

- The chart and table are meant to show the total billing amounts associated with each insurance provider. The table provides exact figures, while the chart offers a visual representation.

2.Sum of Billing Amount by Medical Condition (Waterfall Chart):

- The waterfall chart shows the cumulative billing amounts for various medical conditions, with each green bar representing individual conditions like Cancer, Asthma, and Hypertension. The yellow bar at the end represents the total sum across all conditions.

- Below the chart, a table lists the exact sum of billing amounts for each medical condition:
- **Cancer**: $43,493,080.71
- **Asthma**: $43,412,013.78
- **Hypertension**: $42,534,281.35
- **Diabetes**: $42,295,568.48
- **Obesity**: $41,873,531.89
- **Arthritis**: $41,559,591.57



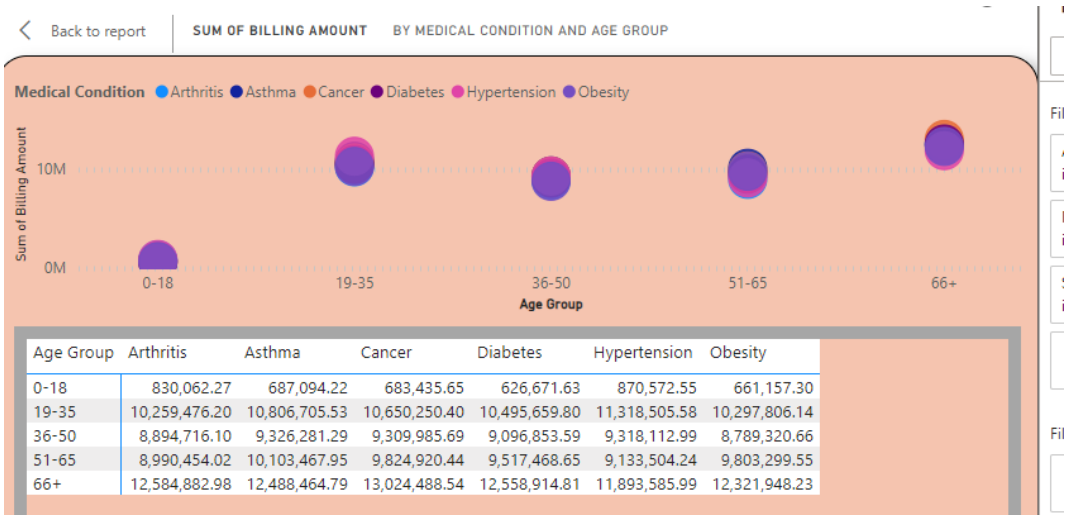
- The waterfall chart effectively shows how each condition contributes to the overall total. The gradual build-up helps in understanding the relative contribution of each condition to the cumulative total.
- The chart is color-coded for clarity, with green indicating individual medical conditions and yellow representing the total.

3. Sum of Billing Amount by Medical Condition and Age Group (Scatter Chart):

- The scatter chart plots the billing amounts by medical condition against different age groups, revealing any correlations between age and treatment costs.

Data Table:

- Below the chart is a table that lists the exact billing amounts for each medical condition across the different age groups. For example:
    - **For Age Group 19-35**:
        - Arthritis: $10,259,476.20
        - Asthma: $10,806,705.53
        - Cancer: $10,650,250.40
        - Diabetes: $11,318,508.53
        - Hypertension: $10,297,806.14
        - Obesity: $10,298,648.96



Age Group	Arthritis	Asthma	Cancer	Diabetes	Hypertension	Obesity
0-18	830,062.27	687,094.22	683,435.65	626,671.63	870,572.55	661,157.30
19-35	10,259,476.20	10,806,705.53	10,650,250.40	10,495,659.80	11,318,505.58	10,297,806.14
36-50	8,894,716.10	9,326,281.29	9,309,985.69	9,096,853.59	9,318,112.99	8,789,320.66
51-65	8,990,454.02	10,103,467.95	9,824,920.44	9,517,468.65	9,133,504.24	9,803,299.55
66+	12,584,882.98	12,488,464.79	13,024,488.54	12,558,914.81	11,893,585.99	12,321,948.23

Decision:

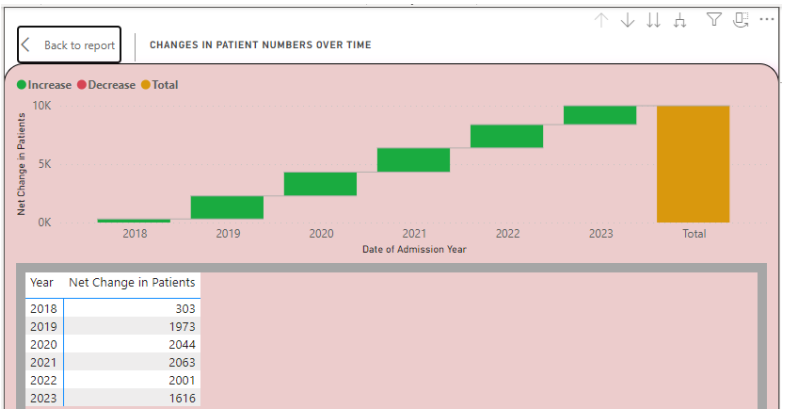
- **Visualization**:

The scatter chart effectively shows the distribution of billing amounts by age group and medical condition. It allows for easy identification of trends, such as which age group has the highest billing for a specific condition.

❖ **Admission and Discharge Data Section:**

1. Net Change in Patient Numbers Over Time (Waterfall Chart):

- Each bar in the waterfall chart represents the yearly net change in patient numbers starting from a base value of 303 patients in 2018. Increases are shown as upward bars, decreases as downward bars, with the final bar indicating the total net change from 2018 to 2023.



Year	Net Change in Patients
2018	303
2019	1973
2020	2044
2021	2063
2022	2001
2023	1616

Yearly Patient Numbers:

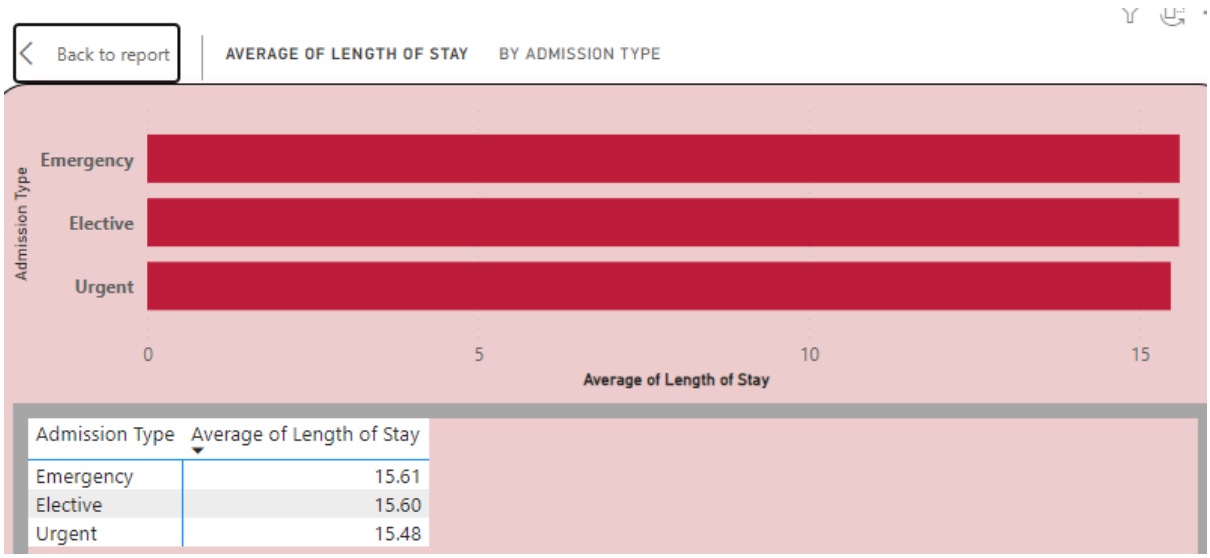
- 2018: 303 (Base Year)
- 2019: 1,973
- 2020: 2,044
- 2021: 2,063
- 2022: 2,001
- 2023: 1,616

Decision:

- **Visualization**:
  - This waterfall chart would effectively demonstrate the trend in patient numbers over the years, highlighting periods of growth (2019 to 2021) and decline (2022 and 2023).
  - The net increase from 2018 to 2023 would also be clearly visible in the final bar, showing the overall growth despite the fluctuations.

2. Average of Length of Stay by Admission Type (Clustered Bar Chart):

- A clustered bar chart was used to compare the average length of stay for different types of admissions (emergency, elective), providing insights into patient management.



Comparison:

- The chart shows that the average length of stay is almost identical across the three admission types, with only slight variations.
- Emergency and Elective admissions have nearly the same average stay, while Urgent admissions have a slightly lower average stay.

Decision:

- **Visualization**:
    - The clustered bar chart effectively demonstrates the minimal difference in the average length of stay across different admission types.

3.Sum of Length of Stay by Gender and Age Group (Scatter Chart):

- This scatter chart visualizes the total length of stay for different gender and age groups, which helps in understanding demographic trends in hospital stays.

Age Group	Female	Male
0-18	1392	1195
19-35	19318	19235
36-50	16945	16202
51-65	17760	17451
66+	23835	22285

Gender Comparison:

- The chart likely shows separate points for males and females within each age group.
- In each age group, females generally have a slightly higher sum of the length of stay compared to males, with the most significant differences in the 66+ age group.

Decision:

- **Visualization**:
    - The scatter chart is effective for comparing the total length of stay between genders across age groups. However, the differences may not be immediately apparent due to the close proximity of the values.

- **Creative and Intuitive Elements**

To enhance dashboard appeal and usability, I incorporated:

- Color Schemes: Consistent palette for visual harmony and category distinction.
- Labels and Tooltips: Clear labels and interactive tooltips for additional context.
- Custom Visuals: Custom visuals like tree maps for engaging categorical data representation.

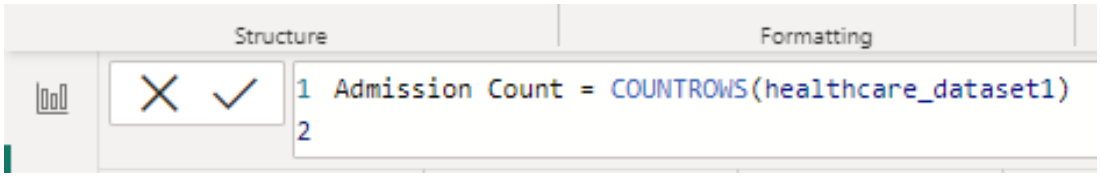
- **Advanced Techniques and Features**

To enhance the dashboards, I applied several advanced techniques in Power BI:

- **DAX Functions:**

  I used DAX functions to create calculated columns and measures for complex calculations, including Admission Count, Age Group, Average Billing Amount, Length of Stay, Max Billing Amounts, and Net Change in Patients.

  - **Admission Count**



- **Age group**

```
Age Group =
SWITCH(
    TRUE(),
    healthcare_dataset1[Age] <= 18, "0-18",
    healthcare_dataset1[Age] <= 35, "19-35",
    healthcare_dataset1[Age] <= 50, "36-50",
    healthcare_dataset1[Age] <= 65, "51-65",
    "66+"
)
```

- **Average Billing Amount**

```
Average Billing Amount = AVERAGE(healthcare_dataset1[Billing Amount])
```

- **Length of stay**

```
Length of Stay = DATEDIFF([Date of Admission], [Discharge Date], DAY)
```

- **Max billing amounts**

```
Max Billing Amount = MAX(healthcare_dataset1[Billing Amount])
```

- **Net Change in Patients**

```
Net Change in Patients =
CALCULATE(
    COUNTROWS(healthcare_dataset1),
    healthcare_dataset1[Date of Admission] - healthcare_dataset1[Discharge Date]
)
```

- **Conditional Formatting:** Conditional formatting was applied to highlight key data points, such as high-cost conditions or extended lengths of stay, making it easier to spot important trends.

- **Drill-Through and Drill-Down:** Drill-through and drill-down features let users explore data from high-level overviews to detailed insights, covering patient demographics, medical conditions, financials, and admission/discharge trends.

Task 4

Evaluation of the Impact of Latest Developments in Data Visualization Technologies

Introduction:

Machine Learning (ML), Artificial Intelligence (AI), Neural Networks (NN), Augmented Reality (AR), Virtual Reality (VR), and Mixed Reality (MR) are key technologies enhancing data visualization in healthcare. This evaluation examines recent advancements and their impact on visualizing healthcare data.

1. Machine Learning (ML):

Latest Developments: ML algorithms have advanced in predictive analytics, pattern recognition, and anomaly detection. Recent research emphasizes the integration of deep learning techniques to improve accuracy in predicting patient outcomes and disease progression.

Impact on Data Visualization: ML enhances visualization by enabling real-time analytics and predictive modeling. It helps in identifying trends, anomalies, and correlations that are crucial for personalized treatment plans and resource allocation.

2. Artificial Intelligence (AI):

Latest Developments: AI technologies, including natural language processing and image recognition, have seen significant advancements. Recent AI systems can analyze unstructured data from electronic health records and medical imaging with high precision.

Impact on Data Visualization: AI-driven visualizations provide deeper insights through automated data analysis and synthesis. This facilitates the creation of interactive and intuitive dashboards that help healthcare professionals make informed decisions.

3. Neural Networks (NN):

Latest Developments: Advances in neural networks, particularly deep learning, have improved their ability to process complex datasets, including medical images and genomic data.

Impact on Data Visualization: Neural networks enhance the visualization of intricate data patterns and structures. They enable the creation of high-resolution visual representations of medical data, improving the interpretation of diagnostic and prognostic information.

4. Augmented Reality (AR):

Latest Developments: AR technology has evolved to offer more immersive and interactive experiences. Recent developments include improved hardware and software that provide real-time overlays of medical data onto physical environments.

Impact on Data Visualization: AR enables the visualization of complex data in real-world contexts, such as overlaying patient information on anatomical models or live images, enhancing surgical planning and patient education.

5. Virtual Reality (VR):

Latest Developments: VR has advanced with more realistic simulations and interactive environments. Research focuses on creating detailed virtual models for training and treatment planning.

Impact on Data Visualization: VR allows immersive data visualization, enabling users to interact with and manipulate 3D medical models. This is especially beneficial for education and detailed exploration of complex datasets.

6. Mixed Reality (MR):

Latest Developments: MR combines elements of AR and VR, offering enhanced interactive experiences. Recent innovations include more sophisticated hardware and software platforms that integrate virtual and real-world elements seamlessly.

Impact on Data Visualization: MR provides a hybrid approach to data visualization, merging real and virtual data elements. This allows for advanced interactive visualizations where users can engage with both digital and physical representations of data, improving situational awareness and decision-making in healthcare.

Conclusion:

Integrating ML, AI, NN, AR, VR, and MR into data visualization is transforming healthcare by providing more accurate, interactive, and immersive data analysis. These technologies improve patient care, streamline processes, and aid decision-making. Ongoing research promises further advancements in visualizing and using healthcare data.

References

Kaggle. (2024). Retrieved from https://www.kaggle.com/